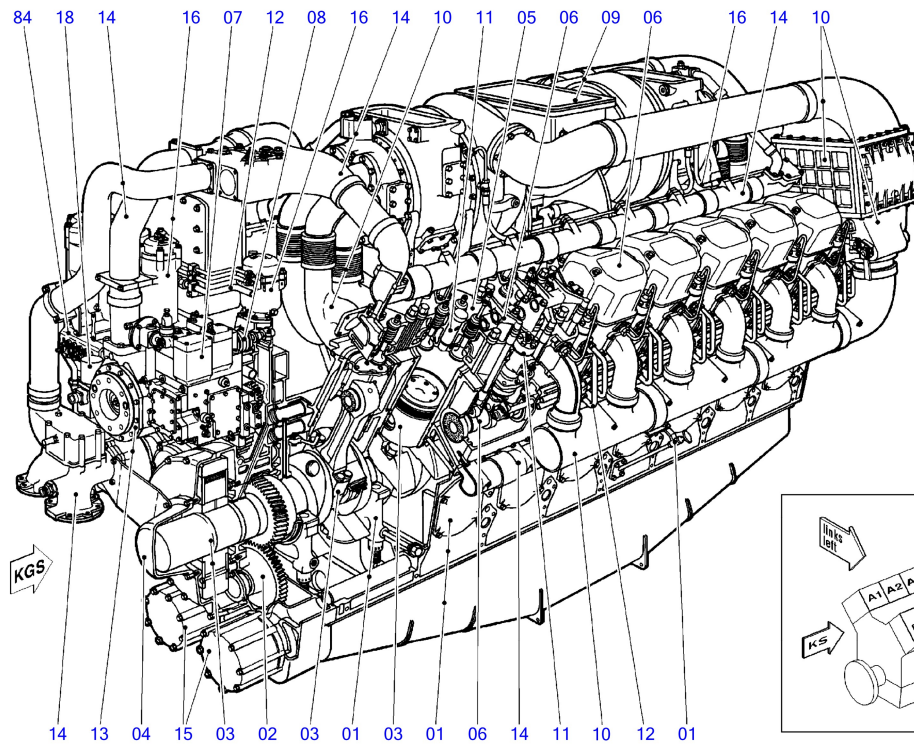
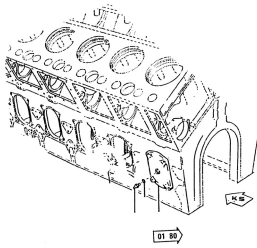




Gilt sinngemäß auch für
 Applies analogously to
 12 V 956 TB .2
 12 V 1163 TB .2
 20 V 956 TB .2
 20 V 1163 TB .2

E 000 5723 a0

- | FG | FG DESIGNATION |
|----|-----------------------------------|
| 01 | CRANKCASE |
| 02 | GEAR TRAIN |
| 03 | RUNNING GEAR |
| 04 | COVER PLATES AT KGS |
| 05 | CYLINDER HEAD |
| 06 | CONTROL SYSTEM |
| 07 | ENGINE GOVERNOR |
| 08 | GOVERNOR LINKAGE |
| 09 | EXHAUST TURBOCHARGER |
| 10 | CHARGE AIR PIPING-EXHAUST PIPING |
| 11 | FUEL INJECTION EQUIPMENT |
| 12 | FUEL SYSTEM |
| 13 | COOLANT PUMP |
| 14 | COOLANT SYSTEM |
| 15 | OIL PUMP |
| 16 | OIL SYSTEM |
| 17 | DRIVE FOR SPEED MONITORING |
| 18 | STARTING EQUIPMENT |
| 19 | ENGINE MOUNTING |
| 84 | MONITORING EQUIPMENT |
| 86 | ACCESSORIES (ELECTRIC, PNEUMATIC) |



mtu M.A.N. Mercedes-Benz Dieselmotor Motoren und Turbinen-Union Friedrichshafen		Name: _____ Adresse: _____ Telefon: _____ Fax: _____ E-Mail: _____
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Hinweise für Betrieb und Wartung	
Motor: Baujahr: Seriennummer: Ölmenge: Ölart: Ölstand:	Betrieb in °C: Last (kW): Drehmoment (Nm): Drehzahl (U/min): Öltemperatur (°C): Ölstand (mm):

Warning

Do not open hot crankcase for at least 10 minutes after shutdown of engine.
Do not restart overheated engine before cause of overheating has been remedied.

Anleitung

ATTENTION: To avoid the consequences of damage, please observe the following instructions:

1. Do not open the hot crankcase for at least 10 minutes after shutdown of the engine.

2. Do not restart the engine before the cause of overheating has been remedied.

Wichtige Hinweise

Only use fluids and lubricants as per MTU specifications. Use only MTU-approved products.

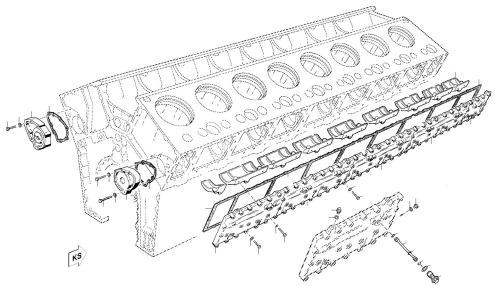
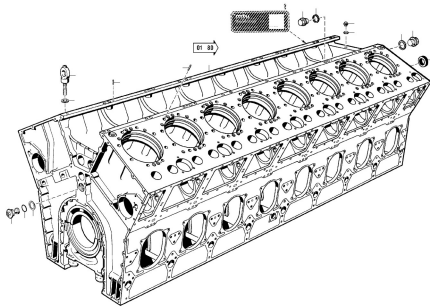
Use only MTU-approved products for lubrication. Use only MTU-approved products for oil.

Use only MTU-approved products for oil. Use only MTU-approved products for oil.

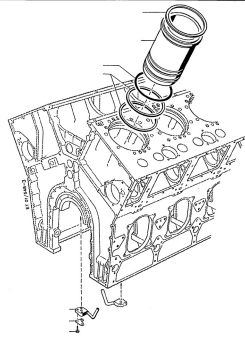
Oil Level

Check the oil level regularly. The oil level should be between the upper and lower marks.

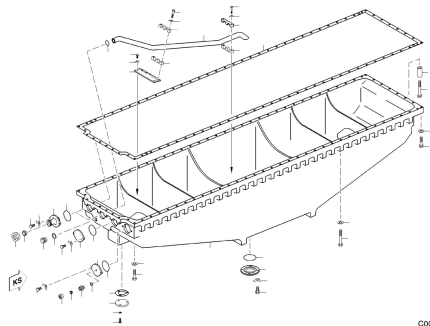
Do not overfill the oil. Overfilling can cause damage to the engine.



01
064003

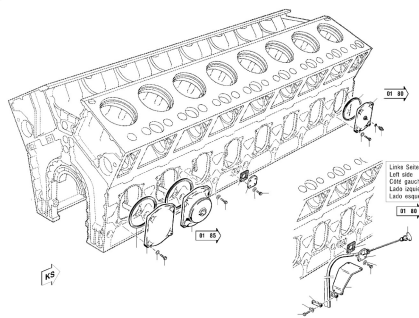


01
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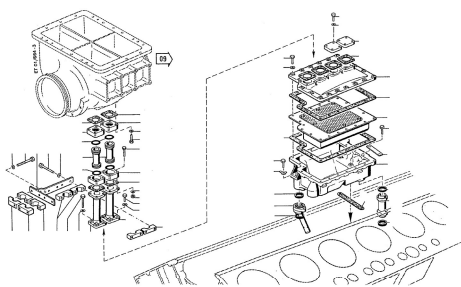
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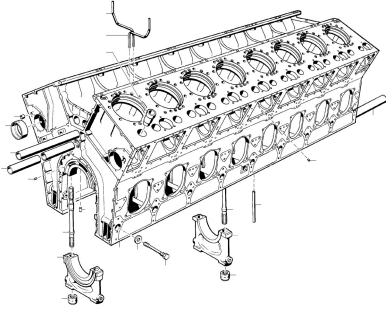


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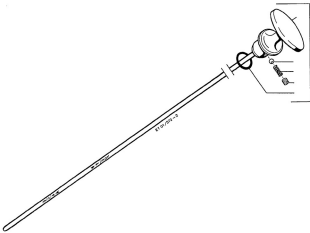


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5288a0

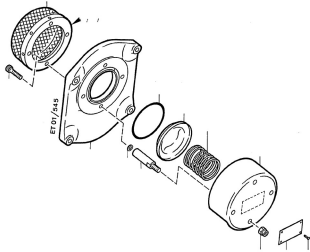


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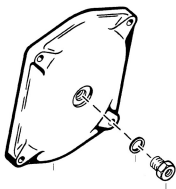
01
037002

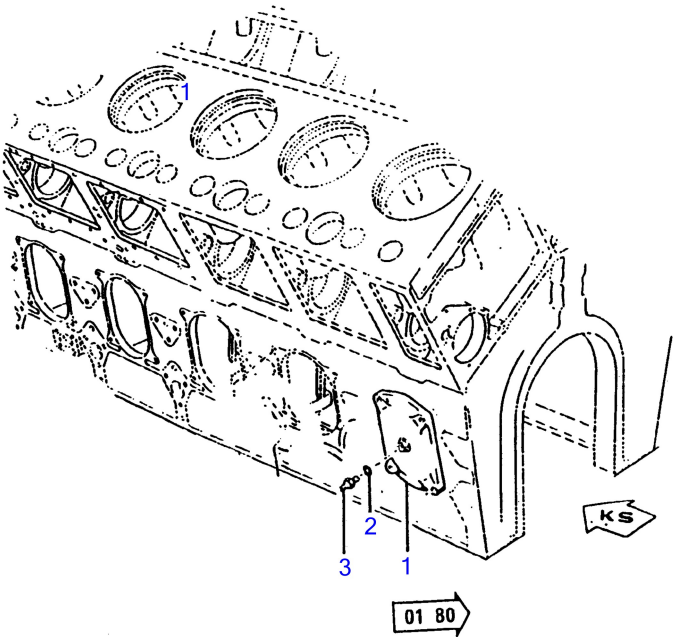


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094500



01
056000





POS.	PART NO.	DESIGNATION	QTY.	UNITS	FN	MEASUREMENT	STANDARD	DES. ADDITION
01	5800100215	INSPECTION PORT COVER	1	EA				
02	007603022102	SEALING RING	1	EA		A 22 X 27		
03	915013016002	EXHAUST PIPE NECK	1	EA		16		

mtu
M.A.N. Maybach Mercedes-Benz
Dieselmotor

Typ _____
Motornr. _____
Baujahr _____
Leistung _____ 1/kW
Drehzahl _____ 1/min
Gewicht _____ kg

Motoren- und Turbinen-Union
Friedrichshafen GmbH

1

Hinweise für Betrieb und Wartung

Motorart _____ Ventilspiel bei [°C] _____
Motornummer _____ Einlaß [mm] _____
Startfüllung _____ Auslaß [mm] _____
Einspritz-Förderbeginn [°C] _____ Vorwärmtemperatur [°C] min. _____
Düsenöffnungsdruck [bar] _____ Öltemperatur [°C] max. _____
Düsentyp _____ Kühlwassertemperatur [°C] max. _____
Zündfolge _____

Alle weiteren technischen Daten sind der Betriebsanleitung zu entnehmen

7

Warning

Do not open hot crankcase for at least 10 minutes after shutdown of engine.

Do not restart overheated engine before cause of overheating has been remedied.

2

Achtung! Motorenöl nur bis zur oberen Markierung am Ölmeßstab einfüllen

ATTENTION! Fill engine oil only up to the upper mark on dipstick

ATTENTION! Ne remplir de l'huile-moteur que jusqu'au repère supérieur sur la jauge d'huile

Atención! Rellenar aceite motor sólo hasta la marcación superior de la varilla indicadora del nivel de aceite

Atenção! Encher óleo de motor só até à marcação superior da vareta medidora do nível de óleo

Attenzione! Introdurre olio motore solo fino alla graduazione superiore dell'asta livello olio

Dikkat! Motor yağına, yağ seviyesi ölçme çubuğunun sadece üst işaretine kadar dokununuz

8

Betriebsstoffe nur nach MTU-Vorschrift Nr. 1061

Only use Fluids and Lubricants as per MTU Specification No. 1061

Utiliser les ingrédients prescrits par Publication MTU No. 1061

Utilizar solamente materias de servicio seg. Publ. MTU No. 1061

Só utilizar agentes de funcionamento seg. Publ. MTU No. 1061

Impiegare solo materiali di consumo sec. Pubbl. MTU No. 1061

YALNIZCA MTU 1061 SARTNAMESİNE UYGUN SIVI VE YAĞLARI KULLANINIZ

Χρησιμοποιείτε μόνο υγρά και λιπαντικά που είναι σύμφωνο με την προδιαγραφή της MTU αρ.1061

Det må kun benyttes brændstoff og smøremidler angitt i MTU-spesifikasjon nr.1061

استخدم فقط المواد ومواد التشحيم المطابقة لمواصفات شركة أم تي يو رقم 1061

3

Vor Tausch des Filtereinsatzes unbedingt Ölschlamm ablassen.

Prior to replacement of filter insert oil sludge must positively be drained.

Prior to replacement of filter insert oil sludge must positively be drained.

Avant de remplacer l'élément filtrant, il faut absolument évacuer les ordures d'huile.

Antes de substituir o elemento filtrante, é indispensável despejar a lama de óleo.

Antes de cambiar el elemento filtrante, es imprescindible evacuar el lodo de aceite.

Prima di sostituire la cartuccia filtrante è assolutamente necessario di scaricare la motina.

SÜZÜCÜ BAĞLAMI DEĞİSTİRİLMEDEN ÖNCE BİRKİNTİ YAĞLAR TAMAMILE BOSALTILMALIDIR.

For inserting av nytt filter må oljeslamm fjernes.

Πριν την αντικατάσταση του φίλτρου αναγκαστικά πρέπει να γίνει απομάκρυνση αποβλήτων λαδιού

قبل تغيير وادجة المرشح يجب تصفية رواسب الزيت تماماً

4

Sach-Nr. _____
Serien-Nr. _____
Nenninhalt _____ dm³
Zul.Betriebsüberdruck _____ bar
Zul.Betriebstemperatur _____ °C

mtu Motoren und Turbinen-Union
Friedrichshafen GmbH

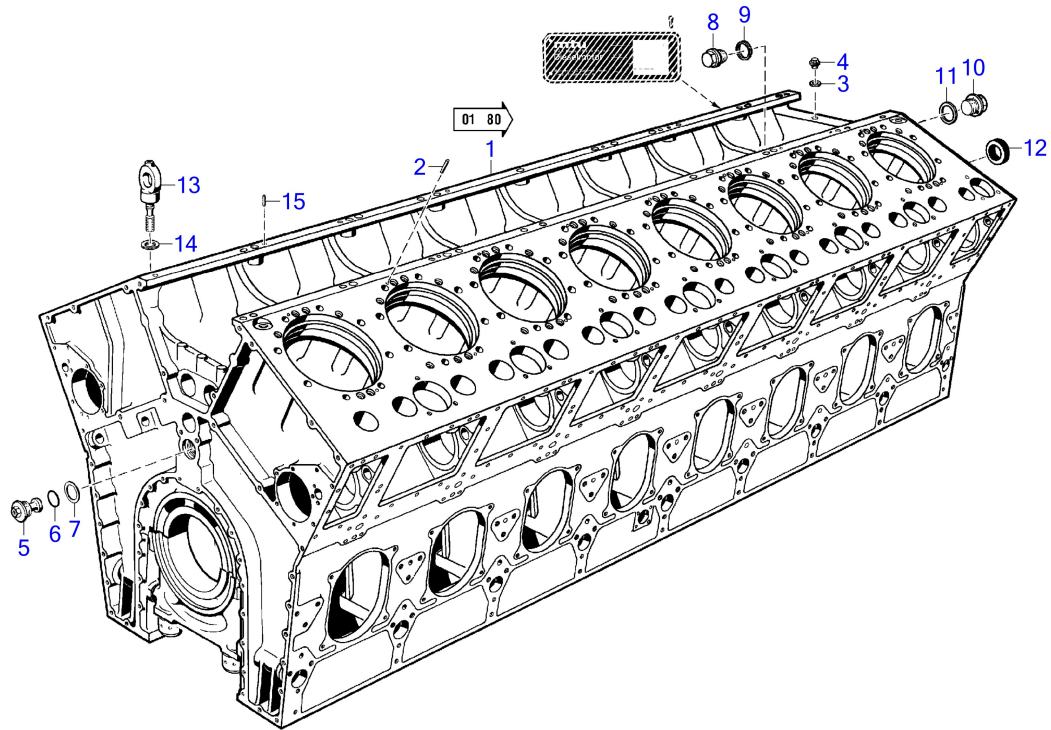
6

14 MMN 425

5

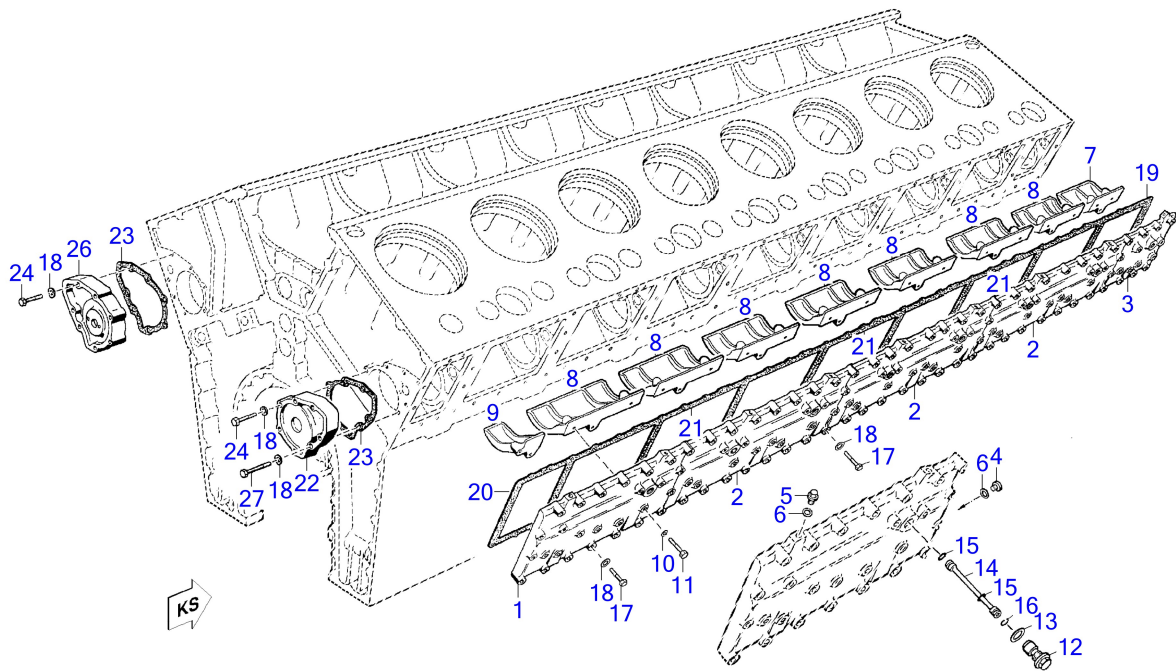
C0010123a0

POS.	PART NO.	DESIGNATION	QTY.	UNITS	FN	MEASUREMENT	STANDARD	DES. ADDITION
01	700425000005	INSTRUCTION PLATE	1	EA		04		
02	700425000039	INSTRUCTION PLATE	1	EA		36		
03	700425000108	INSTRUCTION PLATE	1	EA	NB	40.3		
04	700425000109	INSTRUCTION PLATE	1	EA	NB	41.4		
05	700425000025	INSTRUCTION PLATE	1	EA		14		
06	700425000057	INSTRUCTION PLATE	5	EA		42		
07	700425000073	INSTRUCTION PLATE	1	EA		51		
08	700425000115	INSTRUCTION PLATE	1	EA		61		



C0010033a0

POS.	PART NO.	DESIGNATION	QTY.	UNITS	FN	MEASUREMENT	STANDARD	DES. ADDITION
01	5800108908	CRANKCASE	1	EA				KOMPLETT
02	001481008001	SPRING PIN	32	EA		8 X 20		
03	000910018003	PLUG	1	EA		M 18 X 1,5		
04	007603018101	SEALING RING	1	EA		A 18 X 22		
05	5800110235	PLUG	1	EA				
06	700429048001	O-RING	1	EA		48 X 5		
07	007603064100	SEALING RING	1	EA		A 64 X 72		
08	000910048000	PLUG	1	EA		M 48 X 1,5		
09	007603048101	SEALING RING	1	EA		A 48 X 55		
10	000910052000	PLUG	1	EA		M 52 X 1,5		
11	007603052101	SEALING RING	1	EA		A 52 X 60		
12	5800150560	SEALING RING	2	EA				
13	5802230041	LIFTING EYE	4	EA				
14	5809900240	COVER DISC	4	EA	WW			
14A	5809901040	COVER DISC	4	EA	WW			
14B	5809900940	COVER DISC	4	EA	WW			
14C	5809901340	COVER DISC	4	EA	WW			
15	000007008115	DOWEL PIN	4	EA		8 M6 X 18		



C0010037a0

POS.	PART NO.	DESIGNATION	QTY.	UNITS	FN	MEASUREMENT	STANDARD	DES. ADDITION
01	5800103935	END COVER	2	EA				
02	5800103635	END COVER	6	EA				
03	5800103835	END COVER	2	EA				
04	000908012003	PLUG	8	EA		M 12 X 1,5		
05	000910012000	PLUG	18	EA		M 12 X 1,5		
06	007603012111	SEALING RING	26	EA		A 12 X 15,5		
07	5840100062	OIL COLLECTING TRAY	2	EA				
08	5840100162	OIL COLLECTING TRAY	14	EA				
09	5840100262	OIL COLLECTING TRAY	2	EA				
10	007603010107	SEALING RING	52	EA		A 10 X 16		
11	000933010150	SCREW	52	EA		M 10 X 55		
12	5809970132	PLUG	16	EA				
13	007603027100	SEALING RING	16	EA		A 27 X 32		
14	5800150155	PISTON OIL DUCT	16	EA				
15	700429010000	O-RING	32	EA		10 X 2		
16	007993014200	SNAP RING	16	EA		B 14		
17	000933010095	SCREW	116	EA		M 10 X 50		
18	000125010524	COVER DISC	130	EA		A 10,5		
19	5800110880	GASKET	2	EA				
20	5800110980	GASKET	2	EA				
21	5800111080	GASKET	6	EA				
22	5800152630	END COVER	1	EA				
23	5800152580	GASKET	2	EA				

POS.	PART NO.	DESIGNATION	QTY.	UNITS	FN	MEASUREMENT	STANDARD	DES. ADDITION
24	000933010144	SCREW	13	EA		M 10 X 60		
26	5800102635	END COVER	0	EA	KH			NICHT ANGEBAUT / NOT FITTED
27	000931010239	SCREW	1	EA		M 10 X 80		